IREF - NSF Grant Scenarios

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2025-01-21

Prey Displacement Rate

Part 1: Predator Encounter Rate

For part I, the encounter rate equation is:

$$Z = 2rN * sqrt(v^2 + s^2)$$

where s is prey speed (i.e., displacement rate), v is predator speed, N is predator density, and r is the radius within which the predator can perceive prey.

Scott simulates this equation for varying levels of s, and four constant levels of v (0, 0.5, 1, and 2). Predator density and detection radius are kept constant such that $2rN = 1$.

$v = 0$ would simply mean a sit-and-wait predator.

Part 2: Cost vs. Gain for Displacement Rate

Scott mentions incorporating a Type II functional response, such that higher body velocity has diminishing returns due to time spent handling food.

$$f(R) = \frac{sR}{1 + shR}$$

where s is prey displacement rate, h is handling time, and R is resource density.

Scott describes the fitness curve (without predation, so the growth curve) using the equation from Part 2, and predation using the equation from Part 1.

```r
# Generating the baseline curves
trait.vec <- seq(0, 1, by = 0.01)
speed.vec <- 1 - trait.vec


# These formulas are simplified by setting some of the
# values as constants r is always assumed to be 1, while k,
# s, and v are manipulated values
pred.curve <- function(k, v, s) {
    k * sqrt(v^2 + s^2)
}
```

```r
# R is always assumed to be 1, while s and h are
# manipulated values
growth.curve <- function(s, R, h) {
    (R * s)/(1 + s * R * h)
}


# Build growth curves, for h = 0, 0.5, 1, 2
R_val <- 1

growthcurve.df <- as.data.frame(mat.or.vec(length(speed.vec),
    5))
colnames(growthcurve.df) <- c("s", "h_0", "h_0.5", "h_1", "h_2")
growthcurve.df$s <- trait.vec
growthcurve.df$h_0 <- growth.curve(s = growthcurve.df$s, R = R_val,
    h = 0)
growthcurve.df$h_0.5 <- growth.curve(s = growthcurve.df$s, R = R_val,
    h = 0.5)
growthcurve.df$h_1 <- growth.curve(s = growthcurve.df$s, R = R_val,
    h = 1)
growthcurve.df$h_2 <- growth.curve(s = growthcurve.df$s, R = R_val,
    h = 2)


# Build cost curves, for v = 0, 0.5, 1, 2
k_val <- 1

mortcurve.df <- as.data.frame(mat.or.vec(length(speed.vec), 5))
colnames(mortcurve.df) <- c("s", "v_0", "v_0.5", "v_1", "v_2")
mortcurve.df$s <- trait.vec
mortcurve.df$v_0 <- pred.curve(k = k_val, v = 0, s = mortcurve.df$s)
mortcurve.df$v_0.5 <- pred.curve(k = k_val, v = 0.5, s = mortcurve.df$s)
mortcurve.df$v_1 <- pred.curve(k = k_val, v = 1, s = mortcurve.df$s)
mortcurve.df$v_2 <- pred.curve(k = k_val, v = 2, s = mortcurve.df$s)
```

```r
# Combine the two to generate IREF figures

# This is the baseline scenario
baseline.df <- as.data.frame(mat.or.vec(length(speed.vec), 6))
colnames(baseline.df) <- c("trait", "speed", "growth_curve",
    "mort_curve", "fitness", "NCE_CE")
baseline.df$trait <- trait.vec
baseline.df$speed <- speed.vec
h_val <- 8
R_val <- 1
cost_constant_val <- 9
v_val <- 0.2
k_val <- 0.45
pred_constant_val <- 0.98058

baseline.df$growth_curve <- cost_constant_val * growth.curve(s = baseline.df$speed,
    R = R_val, h = h_val)  # Growth curve
baseline.df$mort_curve <- pred_constant_val * pred.curve(k = k_val,
```

```r
    v = v_val, s = baseline.df$speed)  # Mortality curve
baseline.df$fitness <- baseline.df$growth_curve - baseline.df$mort_curve  # fitness
baseline.df$NCE_CE <- (1 - baseline.df$growth_curve)/baseline.df$mort_curve  # NCE/CE

baseline_optim <- baseline.df$trait[which(baseline.df$fitness ==
    max(baseline.df$fitness))]
baseline_optim_fitness <- baseline.df$fitness[which(baseline.df$fitness ==
    max(baseline.df$fitness))]
baseline_optim_CE <- baseline.df$mort_curve[which(baseline.df$fitness ==
    max(baseline.df$fitness))]
baseline_optim_NCE <- 1 - baseline_optim_CE - baseline_optim_fitness



# Grant figures *appear* to be these params for predator
# density Predator density (k_val) is doubled
preddens.df <- as.data.frame(mat.or.vec(length(speed.vec), 6))
colnames(preddens.df) <- c("trait", "speed", "growth_curve",
    "mort_curve", "fitness", "NCE_CE")
preddens.df$trait <- trait.vec
preddens.df$speed <- speed.vec
h_val <- 8
R_val <- 1
cost_constant_val <- 9
v_val <- 0.2
k_val <- 0.9
pred_constant_val <- 0.98058

preddens.df$growth_curve <- cost_constant_val * growth.curve(s = preddens.df$speed,
    R = R_val, h = h_val)  # Growth curve
preddens.df$mort_curve <- pred_constant_val * pred.curve(k = k_val,
    v = v_val, s = preddens.df$speed)  # Mortality curve
preddens.df$fitness <- preddens.df$growth_curve - preddens.df$mort_curve  # fitness
preddens.df$NCE_CE <- (1 - preddens.df$growth_curve)/preddens.df$mort_curve  # NCE/CE

preddens_optim <- preddens.df$trait[which(preddens.df$fitness ==
    max(preddens.df$fitness))]
preddens_optim_fitness <- preddens.df$fitness[which(preddens.df$fitness ==
    max(preddens.df$fitness))]
preddens_optim_CE <- preddens.df$mort_curve[which(preddens.df$fitness ==
    max(preddens.df$fitness))]
preddens_optim_NCE <- 1 - preddens_optim_CE - preddens_optim_fitness
# Maximum mortality is at 0.8999994


# These *appear* to be the params for foraging mode v is
# five times higher, but k is also (more than) doubled?
# why?
predmode.df <- as.data.frame(mat.or.vec(length(speed.vec), 6))
colnames(predmode.df) <- c("trait", "speed", "growth_curve",
    "mort_curve", "fitness", "NCE_CE")
predmode.df$trait <- trait.vec
predmode.df$speed <- speed.vec
```

```r
h_val <- 8
R_val <- 1
cost_constant_val <- 9
v_val <- 1  # this value changes between the baseline and this one, not sure why
k_val <- 1  # This changes from 0.45, which I think is where the change in predator density is coming f
pred_constant_val <- 0.7071068 * 0.8999994  # this changes to anchor the mortality curve in the same sp


predmode.df$growth_curve <- cost_constant_val * growth.curve(s = predmode.df$speed,
    R = R_val, h = h_val)  # Growth curve
predmode.df$mort_curve <- pred_constant_val * pred.curve(k = k_val,
    v = v_val, s = predmode.df$speed)  # Mortality curve
predmode.df$fitness <- predmode.df$growth_curve - predmode.df$mort_curve  # fitness
predmode.df$NCE_CE <- (1 - predmode.df$growth_curve)/predmode.df$mort_curve  # NCE/CE

predmode_optim <- predmode.df$trait[which(predmode.df$fitness ==
    max(predmode.df$fitness))]
predmode_optim_fitness <- predmode.df$fitness[which(predmode.df$fitness ==
    max(predmode.df$fitness))]
predmode_optim_CE <- predmode.df$mort_curve[which(predmode.df$fitness ==
    max(predmode.df$fitness))]
predmode_optim_NCE <- 1 - predmode_optim_CE - predmode_optim_fitness


# And these *appear* to be the params for resource intake
# rate h is reduced to 1/8 of its original value k remains
# double the baseline, not sure why
prey_h.df <- as.data.frame(mat.or.vec(length(speed.vec), 6))
colnames(prey_h.df) <- c("trait", "speed", "growth_curve", "mort_curve",
    "fitness", "NCE_CE")
prey_h.df$trait <- trait.vec
prey_h.df$speed <- speed.vec
h_val <- 1
R_val <- 1
cost_constant_val <- 2
v_val <- 0.2
k_val <- 0.9
pred_constant_val <- 0.98058

prey_h.df$growth_curve <- cost_constant_val * growth.curve(s = prey_h.df$speed,
    R = R_val, h = h_val)  # Growth curve
prey_h.df$mort_curve <- pred_constant_val * pred.curve(k = k_val,
    v = v_val, s = prey_h.df$speed)  # Mortality curve
prey_h.df$fitness <- prey_h.df$growth_curve - prey_h.df$mort_curve  # fitness
prey_h.df$NCE_CE <- (1 - prey_h.df$growth_curve)/prey_h.df$mort_curve  # NCE/CE

prey_h_optim <- prey_h.df$trait[which(prey_h.df$fitness == max(prey_h.df$fitness))]
prey_h_optim_fitness <- prey_h.df$fitness[which(prey_h.df$fitness ==
    max(prey_h.df$fitness))]
prey_h_optim_CE <- prey_h.df$mort_curve[which(prey_h.df$fitness ==
    max(prey_h.df$fitness))]
prey_h_optim_NCE <- 1 - prey_h_optim_CE - prey_h_optim_fitness
```

```r
# Fig. 1
fig1_a.plot <- ggplot(data = baseline.df, aes(x = trait)) + geom_smooth(aes(y = growth_curve),
    se = FALSE, color = "blue") + geom_smooth(aes(y = mort_curve),
    se = FALSE, color = "red", linetype = "dashed") + theme_classic(base_size = 10,
    base_line_size = 1) + ylab("Growth and Mortality") + xlab("Displacement Rate") +
    scale_x_continuous(expand = c(0, 0), breaks = c(0, 0.5, 1),
        labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + annotate("text",
    x = 0.25, y = 0.85, label = "Growth", size = 8/.pt) + annotate("text",
    x = 0.5, y = 0.35, label = "Mortality", size = 8/.pt) + theme(panel.border = element_rect(color = "
    fill = NA, linewidth = 0.75), aspect.ratio = 1)

fig1_b.plot <- ggplot(data = baseline.df, aes(x = trait)) + geom_smooth(aes(y = growth_curve,
    color = "Growth"), se = FALSE) + geom_smooth(aes(y = mort_curve,
    color = "Mortality"), se = FALSE, linetype = "dashed") +
    geom_smooth(aes(y = fitness, color = "Fitness"), se = FALSE) +
    theme_classic(base_size = 10, base_line_size = 1) + ylab("Growth and Mortality") +
    xlab("Displacement Rate") + geom_vline(xintercept = baseline_optim,
    size = 0.5, linetype = "dashed", colour = "green") + annotate("text",
    x = 0.3, y = 0.7, label = "Fitness", size = 8/.pt) + annotate("text",
    x = 0.35, y = 0.1, label = "Optimal Trait", size = 8/.pt) +
    scale_x_continuous(expand = c(0, 0), breaks = c(0, 0.5, 1),
        labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + scale_colour_manual(name = "Curve",
    values = c("green", "blue", "red")) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1, axis.title.y = element_blank(),
    legend.position = "bottom")

fig1_c.plot <- ggplot(data = baseline.df, aes(x = trait)) + geom_smooth(aes(y = growth_curve),
    color = "blue", se = FALSE) + geom_smooth(aes(y = mort_curve),
    color = "red", se = FALSE, linetype = "dashed") + theme_classic(base_size = 10,
    base_line_size = 1) + ylab("Growth and Mortality") + xlab("Displacement Rate") +
    annotate(geom = "rect", xmin = baseline_optim - 0.025, xmax = baseline_optim +
        0.025, ymin = 1 - baseline_optim_NCE, ymax = 0.998, color = "gray2",
        fill = "lightblue", alpha = 0.5) + annotate("text", x = baseline_optim +
    0.05, y = 1 - baseline_optim_NCE/2, label = "NCE = 0.12",
    hjust = "left", size = 7/.pt) + annotate(geom = "rect", xmin = baseline_optim -
    0.025, xmax = baseline_optim + 0.025, ymin = baseline_optim_CE,
    ymax = baseline_optim_CE + baseline_optim_fitness, color = "gray2",
    fill = "lightgreen", alpha = 0.5) + annotate("text", x = baseline_optim +
    0.05, y = baseline_optim_CE + baseline_optim_fitness/2, label = "0.66",
    hjust = "left", size = 7/.pt) + annotate(geom = "rect", xmin = baseline_optim -
    0.025, xmax = baseline_optim + 0.025, ymin = 0.001, ymax = baseline_optim_CE,
    color = "gray2", fill = "red", alpha = 0.5) + annotate("text",
    x = baseline_optim + 0.05, y = baseline_optim_CE/2, label = "CE = 0.22",
    hjust = "left", size = 7/.pt) + scale_x_continuous(expand = c(0,
    0), breaks = c(0, 0.5, 1), labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1, axis.title.y = element_blank())



get_legend_35 <- function(plot) {
```

```r
    # return all legend candidates
    legends <- get_plot_component(plot, "guide-box", return_all = TRUE)
    # find non-zero legends
    nonzero <- vapply(legends, \(x) !inherits(x, "zeroGrob"),
        TRUE)
    idx <- which(nonzero)
    # return first non-zero legend if exists, and otherwise
    # first element (which will be a zeroGrob)
    if (length(idx) > 0) {
        return(legends[[idx[1]]])
    } else {
        return(legends[[1]])
    }
}

fig1.legend <- get_legend_35(fig1_b.plot)
fig1_b.plot <- fig1_b.plot + theme(legend.position = "none")

fig1.plot <- plot_grid(fig1_a.plot, fig1_b.plot, fig1_c.plot,
    ncol = 3, labels = "auto", label_y = 0.75, align = c("hv"))
fig1.plot <- plot_grid(fig1.plot, fig1.legend, rel_heights = c(1,
    0.1), ncol = 1)

# ggsave('fig1.png', plot = fig1.plot, dpi = 300)
```

```r
# Fig. 2 - predator density
preddens.df <- cbind(preddens.df, baseline.df$mort_curve)
colnames(preddens.df) <- c("trait", "speed", "growth_curve",
    "mort_curve", "fitness", "NCE_CE", "mort_baseline")

preddens_1.plot <- ggplot(data = preddens.df, aes(x = trait)) +
    geom_line(aes(y = growth_curve), color = "blue") + geom_line(aes(y = mort_curve),
    color = "red", linetype = "solid") + geom_line(aes(y = mort_baseline),
    color = "red", linetype = "dashed") + geom_ribbon(aes(ymin = mort_curve,
    ymax = mort_baseline), fill = "red", alpha = 0.25) + geom_segment(aes(x = 0.5,
    y = 0.3, xend = 0.5, yend = 0.45), arrow = arrow(length = unit(0.5,
    "cm")), color = "red", size = 1) + geom_vline(xintercept = baseline_optim,
    size = 0.5, linetype = "dashed", colour = "green") + theme_classic(base_size = 10,
    base_line_size = 1) + ylab("Growth and Mortality") + xlab("Displacement Rate") +
    scale_x_continuous(expand = c(0, 0), breaks = c(0, 0.5, 1),
        labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1, axis.title.x = element_blank(),
    axis.text.x = element_blank())

preddens_2.plot <- ggplot(data = preddens.df, aes(x = trait)) +
    geom_line(aes(y = growth_curve, colour = "Growth"), ) + geom_line(aes(y = mort_curve,
    colour = "Mortality"), linetype = "solid") + annotate("rect",
    xmin = baseline_optim, xmax = preddens_optim, ymin = 0, ymax = 1,
    fill = "green", alpha = 0.25) + theme_classic(base_size = 10,
    base_line_size = 1) + ylab("Growth and Mortality") + xlab("Displacement Rate") +
    geom_vline(xintercept = baseline_optim, size = 0.5, linetype = "dashed",
        colour = "green") + geom_segment(aes(x = baseline_optim +
```

```r
    0.01, y = 0.25, xend = preddens_optim - 0.01, yend = 0.25),
    arrow = arrow(length = unit(0.5, "cm")), color = "green",
    size = 1) + geom_vline(xintercept = preddens_optim, size = 0.5,
    linetype = "solid", colour = "green") + annotate("text",
    x = preddens_optim + 0.15, y = 0.95, label = "NCE = 0.20",
    size = 7/.pt) + annotate("text", x = preddens_optim + 0.15,
    y = 0.45, label = "0.48", size = 7/.pt) + annotate("text",
    x = preddens_optim + 0.15, y = 0.15, label = "CE = 0.32",
    size = 7/.pt) + scale_x_continuous(expand = c(0, 0), breaks = c(0,
    0.5, 1), labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + scale_colour_manual(name = "Curve",
    values = c("blue", "red")) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1) + theme(legend.position = "bottom")

preddens.legend <- get_legend_35(preddens_2.plot)
preddens_2.plot <- preddens_2.plot + theme(legend.position = "none")

preddens.fig <- plot_grid(preddens_1.plot, preddens_2.plot, ncol = 1,
    labels = "auto", hjust = -12.5, align = c("hv"))
preddens.fig <- plot_grid(preddens.fig, preddens.legend, ncol = 1,
    rel_heights = c(1, 0.1))

# ggsave('fig2.png', plot = preddens.fig, dpi = 300)
```

```r
# Fig. 4 - predator hunting mode
predmode.df <- cbind(predmode.df, preddens.df$mort_curve)
colnames(predmode.df) <- c("trait", "speed", "growth_curve",
    "mort_curve", "fitness", "NCE_CE", "mort_baseline")

predmode_1.plot <- ggplot(data = predmode.df, aes(x = trait)) +
    geom_line(aes(y = growth_curve), color = "blue") + geom_line(aes(y = mort_curve),
    color = "red", linetype = "solid") + geom_line(aes(y = mort_baseline),
    color = "red", linetype = "dashed") + geom_ribbon(aes(ymin = mort_curve,
    ymax = mort_baseline), fill = "red", alpha = 0.25) + geom_vline(xintercept = preddens_optim,
    size = 0.5, linetype = "dashed", colour = "green") + geom_segment(aes(x = 0.5,
    y = 0.5, xend = 0.5, yend = 0.7), arrow = arrow(length = unit(0.5,
    "cm")), color = "red", size = 1) + theme_classic(base_size = 10,
    base_line_size = 1) + ylab("Growth and Mortality") + xlab("Displacement Rate") +
    scale_x_continuous(expand = c(0, 0), breaks = c(0, 0.5, 1),
        labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1, axis.title.x = element_blank(),
    axis.text.x = element_blank())

predmode_2.plot <- ggplot(data = predmode.df, aes(x = trait)) +
    geom_line(aes(y = growth_curve, colour = "Growth"), ) + geom_line(aes(y = mort_curve,
    colour = "Mortality"), linetype = "solid") + annotate("rect",
    xmin = preddens_optim, xmax = predmode_optim, ymin = 0, ymax = 1,
    fill = "green", alpha = 0.25) + theme_classic(base_size = 10,
    base_line_size = 1) + ylab("Growth and Mortality") + xlab("Displacement Rate") +
    geom_vline(xintercept = preddens_optim, size = 0.5, linetype = "dashed",
        colour = "green") + geom_vline(xintercept = predmode_optim,
    size = 0.5, linetype = "solid", colour = "green") + annotate("text",
```

```r
    x = predmode_optim, y = 0.975, label = "NCE = 0.09", size = 7/.pt) +
    annotate("text", x = predmode_optim - 0.125, y = 0.85, label = "0.11",
        size = 7/.pt) + annotate("text", x = predmode_optim -
    0.125, y = 0.5, label = "CE = 0.80", size = 7/.pt) + geom_segment(aes(x = preddens_optim -
    0.01, y = 0.25, xend = predmode_optim + 0.01, yend = 0.25),
    arrow = arrow(length = unit(0.5, "cm")), color = "green",
    size = 1) + scale_x_continuous(expand = c(0, 0), breaks = c(0,
    0.5, 1), labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + scale_colour_manual(name = "Curve",
    values = c("blue", "red")) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1) + theme(legend.position = "bottom")

predmode.legend <- get_legend_35(predmode_2.plot)
predmode_2.plot <- predmode_2.plot + theme(legend.position = "none")

predmode.fig <- plot_grid(predmode_1.plot, predmode_2.plot, ncol = 1,
    labels = "auto", hjust = -12.5, align = c("hv"))
predmode.fig <- plot_grid(predmode.fig, predmode.legend, ncol = 1,
    rel_heights = c(1, 0.1))

# ggsave('fig4.png', plot = predmode.fig, dpi = 300)
```

```r
prey_h.df <- cbind(prey_h.df, baseline.df$growth_curve)
colnames(prey_h.df) <- c("trait", "speed", "growth_curve", "mort_curve",
    "fitness", "NCE_CE", "growth_baseline")

# Fig. 4 - Prey handling time
prey_h_1.plot <- ggplot(data = prey_h.df, aes(x = trait)) + geom_line(aes(y = growth_curve),
    color = "blue") + geom_line(aes(y = mort_curve), color = "red",
    linetype = "solid") + geom_line(aes(y = growth_baseline),
    color = "blue", linetype = "dashed") + geom_vline(xintercept = preddens_optim,
    size = 0.5, linetype = "dashed", colour = "green") + geom_ribbon(aes(ymin = growth_baseline,
    ymax = growth_curve), fill = "blue", alpha = 0.25) + geom_segment(aes(x = 0.5,
    y = 0.85, xend = 0.5, yend = 0.7), arrow = arrow(length = unit(0.5,
    "cm")), color = "blue", size = 1) + theme_classic(base_size = 10,
    base_line_size = 1) + ylab("Growth and Mortality") + xlab("Displacement Rate") +
    scale_x_continuous(expand = c(0, 0), breaks = c(0, 0.5, 1),
        labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1, axis.title.x = element_blank(),
    axis.text.x = element_blank())

prey_h_2.plot <- ggplot(data = prey_h.df, aes(x = trait)) + geom_line(aes(y = growth_curve,
    colour = "Growth"), ) + geom_line(aes(y = mort_curve, colour = "Mortality"),
    linetype = "solid") + annotate("rect", xmin = preddens_optim,
    xmax = prey_h_optim, ymin = 0, ymax = 1, fill = "green",
    alpha = 0.25) + theme_classic(base_size = 10, base_line_size = 1) +
    ylab("Growth and Mortality") + xlab("Displacement Rate") +
    geom_vline(xintercept = preddens_optim, size = 0.5, linetype = "dashed",
        colour = "green") + geom_vline(xintercept = prey_h_optim,
    size = 0.5, linetype = "solid", colour = "green") + geom_segment(aes(x = preddens_optim -
    0.05, y = 0.25, xend = prey_h_optim + 0.01, yend = 0.25),
    arrow = arrow(length = unit(0.5, "cm")), color = "green",
```

```
    size = 1) + annotate("text", x = prey_h_optim - 0.15, y = 0.925,
    label = "NCE = 0.29", size = 7/.pt) + annotate("text", x = prey_h_optim -
    0.15, y = 0.75, label = "0.19", size = 7/.pt) + annotate("text",
    x = prey_h_optim - 0.15, y = 0.35, label = "CE = 0.52", size = 7/.pt) +
    scale_x_continuous(expand = c(0, 0), breaks = c(0, 0.5, 1),
        labels = c(1000, 500, 1)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + scale_colour_manual(name = "Curve",
    values = c("blue", "red")) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1) + theme(legend.position = "bottom")

prey_h.legend <- get_legend_35(prey_h_2.plot)
prey_h_2.plot <- prey_h_2.plot + theme(legend.position = "none")

prey_h.fig <- plot_grid(prey_h_1.plot, prey_h_2.plot, ncol = 1,
    labels = "auto", hjust = -12.5, align = c("hv"))
prey_h.fig <- plot_grid(prey_h.fig, prey_h.legend, ncol = 1,
    rel_heights = c(1, 0.1))

ggsave("fig5.png", plot = prey_h.fig, dpi = 300)
```

Predator Density Effects

For this last section, I'd like to make a figure illustrating the very simple, intuitive point that below a certain predator density, the optimal strategy is actually to not respond at all.